The enlargement in Remark 4.2 is not even order-three c-monotone

Automated mathematics discovery run `fa_banach_001`
Lane 13, model GPT5.6

9 August 2026

Status. Candidate counterexample, likely valid, subject to human review. It fully answers the explicit question whether the particular enlargement $\widetilde{\Gamma} = \Gamma \cup \{(0,0,0)\}$ is c-cyclically monotone. It does not decide whether Γ has some other proper c-cyclically monotone extension.

1 Source question

The source is Tongseok Lim, *Maximal Monotonicity and Cyclic Involutivity of Multiconjugate Convex Functions*, arXiv:2207.04830, published in *SIAM Journal on Optimization* 33(4) (2023), 2489–2511, DOI 10.1137/23M1549201. Example 4.1 constructs a contact set $\Gamma \subset (\mathbb{R}^2)^3$ and proves that

$$\widetilde{\Gamma} := \Gamma \cup \{(0,0,0)\}$$

is c-monotone. Remark 4.2 on PDF page 26 conjectures that $\widetilde{\Gamma}$ is c-cyclically monotone and leaves its verification open.

26 TONGSEOK LIM

Remark 4.2. *The author conjectures that $\tilde{\Gamma}$ is also c-cyclically monotone, and thus Γ is not maximally c-cyclically monotone either. However, verifying the c-cyclically monotonicity of $\tilde{\Gamma}$ appears to be nontrivial (despite the fact that we have a complete description of Γ!), and we leave it as an open question.*

Figure 1: The open question in Remark 4.2 of the source paper.

For three marginals the cost is

$$c(x,y,z) = \langle x,y\rangle + \langle y,z\rangle + \langle z,x\rangle.$$

A set is order-three c-monotone if, for every three of its points and every independent permutation of the three coordinate lists, the permuted total cost is no larger than the original total cost. Failure at order three already rules out c-cyclical monotonicity.

2 Proof intuition

The source example takes $x = au$ and $y = bv$ on two oblique line segments. For a fixed z, membership in the contact set says exactly that (a, b) maximizes a bilinear-affine function on a square. At two special, opposite choices of z, that function is flat along an entire edge. We choose one interior point from each flat edge.

Now adjoin the origin, swap only the origin's x-coordinate with the first contact point, and swap only its y-coordinate with the second. The two contact values survive as one-coordinate pairings with their original z's, while the origin acquires a new positive cross term. The permuted sum therefore exceeds the original sum by $\lambda^2/8$.

3 Explicit counterexample

Retain all notation of Example 4.1. Thus $\lambda > 0$,

$$u = (1, 0), \qquad v = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad \langle u, v\rangle = \frac{1}{2},$$

and Γ is the contact set generated by the functions f, g, h there. Put

$$w = v - u = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

and define

$$\begin{aligned} P_0 &= (0, 0, 0), \\ P_1 &= \left(-\frac{\lambda}{2}u,\ \lambda v,\ \lambda w\right), \\ P_2 &= \left(\lambda u,\ -\frac{\lambda}{2}v,\ -\lambda w\right). \end{aligned}$$

Theorem 1. *The points P_1, P_2 belong to Γ, but the three points $P_0, P_1, P_2 \in \widetilde{\Gamma}$ violate the order-three c-monotonicity inequality. More precisely, a coordinate permutation increases the total cost by $\lambda^2/8$. Consequently $\widetilde{\Gamma}$ is not 3-c-monotone and hence is not c-cyclically monotone.*

Proof. We first verify membership without relying only on the region diagram in the source. Write $x = a\lambda u$, $y = b\lambda v$, where $-1 \le a, b \le 1$. Step 1 of Example 4.1 gives

$$h(z) = \sup_{|a|,|b|\le 1} \big(c(a\lambda u, b\lambda v, z)\big).$$

Thus $(a\lambda u, b\lambda v, z) \in \Gamma$ precisely when (a, b) attains this supremum.

For $z = \lambda w$, using $\langle w, u\rangle = -1/2$ and $\langle w, v\rangle = 1/2$, we obtain

$$\lambda^{-2} c(a\lambda u, b\lambda v, \lambda w) = \frac{ab}{2} - \frac{a}{2} + \frac{b}{2}.$$

For fixed $a \in [-1, 1]$, the coefficient of b is $(a + 1)/2 \ge 0$. Hence the maximum occurs at $b = 1$, where the value is exactly $1/2$ for every a. Taking $a = -1/2$ proves $P_1 \in \Gamma$.

Likewise, for $z = -\lambda w$,

$$\lambda^{-2} c(a\lambda u, b\lambda v, -\lambda w) = \frac{ab}{2} + \frac{a}{2} - \frac{b}{2}.$$

For fixed b, the coefficient of a is $(b+1)/2 \geq 0$, so the maximum occurs at $a = 1$, again with value $1/2$ for every b. Taking $b = -1/2$ proves $P_2 \in \Gamma$. In particular,

$$c(P_0) = 0, \qquad c(P_1) = c(P_2) = \frac{\lambda^2}{2},$$

and the original total cost is λ^2.

Index the points by $0, 1, 2$. Let $\sigma_x = (0\ 1)$, $\sigma_y = (0\ 2)$, and $\sigma_z = \mathrm{id}$. The three permuted tuples are

$$(x_1, y_2, z_0), \qquad (x_0, y_1, z_1), \qquad (x_2, y_0, z_2).$$

Their costs are respectively

$$\left\langle -\frac{\lambda}{2} u, -\frac{\lambda}{2} v \right\rangle = \frac{\lambda^2}{8}, \qquad \langle \lambda v, \lambda w \rangle = \frac{\lambda^2}{2}, \qquad \langle \lambda u, -\lambda w \rangle = \frac{\lambda^2}{2}.$$

The permuted total is therefore

$$\frac{9\lambda^2}{8} > \lambda^2,$$

contrary to the defining order-three c-monotonicity inequality. □

4 Verification notes

The exact checker in `code/verify_certificate.py` works in the scaled coordinates

$$(a, b, \alpha, \beta), \qquad \lambda^{-2} c = \frac{ab}{2} + a\alpha + b\beta,$$

using rational arithmetic. It verifies that both selected pairs maximize over all four vertices of the square, that the original normalized sum is 1, and that the permuted sum is 9/8. The accompanying randomized search is retained only as discovery provenance; it is not used in the proof.

Independent hand checks recommended for review are: (i) confirm the cost inequality direction in the definition on source PDF pages 3–4; (ii) confirm the contact-set supremum formula in Step 1 of Example 4.1; and (iii) check the two permutations and three displayed permuted costs.

5 Scope and novelty check

This is a counterexample to the particular conjecture in Remark 4.2. It does not prove that Γ is maximally c-cyclically monotone, nor does it exclude a different proper cyclically monotone extension of Γ.

A bounded novelty search was performed on 9 August 2026. It included the run's registry, solution, attempt, and proof-gap indexes; an exact-phrase and keyword search of the loaded arXiv source corpus; web searches for the exact Remark 4.2 wording, the paper title together with the conjecture, and citing work; the current arXiv record; the published SIAM page; and the author-hosted paper. No later proof or counterexample for this exact remark was located. The author-hosted/current paper still presents it as open. This is a bounded search rather than a comprehensive citation review, so the novelty assessment is *plausibly new, moderate confidence.*

6 Human-review recommendation

Recommend priority review and author notification after validation. The certificate is only three points and one pair of transpositions, uses no external theorem beyond the source's definition of its own contact set, and has an exact positive margin $\lambda^2/8$. These features make the claim especially amenable to rapid independent verification.

References

[1] T. Lim, *Maximal Monotonicity and Cyclic Involutivity of Multiconjugate Convex Functions*, SIAM J. Optim. **33** (2023), no. 4, 2489–2511; arXiv:2207.04830; DOI: 10.1137/23M1549201.